Division properties of exponents and [redacted] zero exponent

1 Simplify the following, giving your answers in exponential form.

a $10^8 \div 10^6$ b $j^8 \div j^4$ c $a^5 \div a^3$ d $9q^6 \div \left(4q^2\right)$

2 Simplify the following, giving your answers with positive exponents.

a $\dfrac{18^{32}}{18^{23}}$ b $\dfrac{a^{10}}{a^6}$ c $\dfrac{u^{13}}{u^2}$ d $\dfrac{t^7r^8}{t^5r^4}$

e $\dfrac{5r^7}{r^4}$ f $\dfrac{3j^{13}}{4j^8}$ g $\dfrac{-x^{11}}{3x^4}$ h $\dfrac{9j^9k^8}{4j^7}$

i $\dfrac{3g^8}{2g^5h^6}$ j $\dfrac{9j^5k^7}{j^3k^4}$ k $\dfrac{j^8k^9}{7j^7k^8}$ l $\dfrac{3m^9n^4}{8m^8n^2}$

m $\dfrac{4p^3q^9}{2p^2q^7}$ n $\dfrac{2p^9q^6}{6p^3q^3}$ o $\dfrac{3j^6k^4}{2j^4k^2}$ p $\dfrac{y^6}{(4y)^2}$

q $\dfrac{6r^3s^6}{2r^2s^4}$ r $\dfrac{-9x^{13}}{3x^4}$ s $\dfrac{x^6}{4x^4}$ t $\dfrac{54x^{11}}{6x^6}$

3 For the following, write the integer value or the term that should go in the space.

a $a^9 \div a^{\square} = a^7$

b $15j^{14} \div (\square) = 5j^7$

c $x^{23}y^{10} \div (\square) = x^{15}y^5$

d $\square \div \left(g^4h^3\right) = g^{16}h^5$

e $63x^{18} \div \square = 9x^{14}$

f $x^{21}y^8 \div \square = x^{14}y^3$

g $\dfrac{8r^3s^8}{\square} = 2rs^2$

4 Simplify the following, giving your answers in exponential form:

a $\dfrac{\left(n^8r^5\right)^5}{\left(n^4r\right)^5}$

b $\dfrac{30x^{13}y^{26}z^{21}}{3x^{11}y^{15}z^6}$

c $\dfrac{15x^{16}y^{25}z^{13}}{-3x^{14}y^{11}z^2}$

d $\left(m^{12}\right)^9 \div \left(m^4\right)^2$

e $\left(-240u^{32}\right) \div \left(-8u^9\right) \div \left(-5u^{12}\right)$

f $\dfrac{y^7 \cdot y^5}{y^3 \cdot y^2}$

g $\dfrac{10p^6 \cdot 3p^{10}}{15p^2}$

h $\dfrac{\left(2x^4y^0\right)^2}{x^6}$

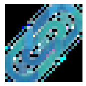
i $\dfrac{6\left(w^2v\right)^5 \cdot 27y^{17}v^7}{\left(3w^5v^3\right)^2 \cdot 2y^6v}$

j $m^9 \div m^5 \cdot m^4$

k $p^{18} \div p^8 \div p^5$

l $\dfrac{\left(x^3\right)^2}{\ }$

m $\dfrac{45b^6}{(3b)(5a)}$

 $\dfrac{210p^{22}}{7p^8} \div \left(5p^7\right)$

5 Fill in the missing value to complete the pattern.

$k^5 = 1 \cdot k \cdot k \cdot k \cdot k \cdot k$

$k^4 = 1 \cdot k \cdot k \cdot k \cdot k$

$k^3 = 1 \cdot k \cdot k \cdot k$

$k^2 = 1 \cdot k \cdot k$

$k^1 = 1 \cdot k$

$k^0 = \square$

6 Fill in the missing exponent to make the following equations true.

a $p^{12} \div p^{12} = p^{\square}$

b $\dfrac{b^{\square}}{b^{11}} = b^0$

c $\dfrac{w^7}{w^{\square}} = w^0$

7 Evaluate:

a 741^0

b $(-983)^0$

c $(2 \cdot 13)^0$

8 Simplify:

a $18a^0$

b q^0

c $11a^0$

d $(6a)^0$

e $\left(a^0\right)^{79}$

f $9 \cdot \left(15x^6\right)^0$

g $\left(3m^4\right)^3 \cdot mq^0$

h $\dfrac{45x^7y^7}{9x^7y^2}$

Additional questions

9 When simplifying the expression $\dfrac{a^5 \cdot a^4}{a^2}$, Chad's first line of work was $a^3 \cdot a^2$.
Is Chad's work correct? Explain.

10 When simplifying the expression $\dfrac{a^5 + a^4}{a^2}$, Clarice's first line of work was $a^3 + a^2$.
Is Clarice's work correct? Explain.